All About Aviation (Airplanes) and Aerodynamics

Made by: Adarsh Sheth

COMMERCIAL

GENERAL AVIATION

3 Principals of Flight

1 Float

Be lighter than the air (i.e. hot air balloons and blimps)

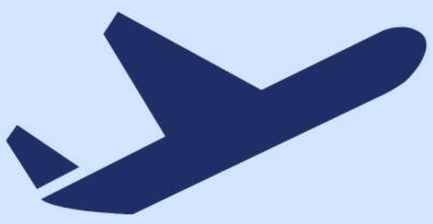
2 Push Air Down

Through the use of airfoils which generate lift (i.e. planes and helicopters)

3 Push Something Else Down

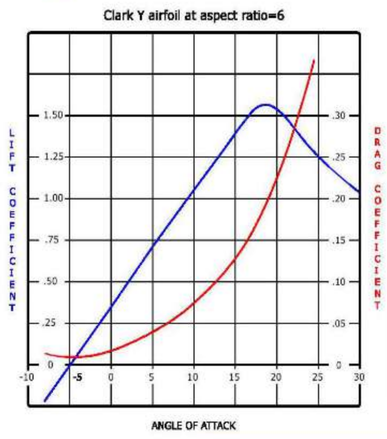
An object such as a rocket which pushes down hot air (by mixing then igniting) fuel and air) to lift itself up (Newton's 3rd Law)

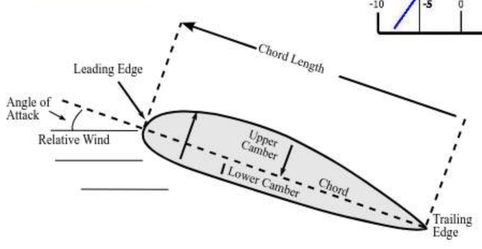
Commercial airplane wings need to be very flexible. Here is a picture of a wing test!

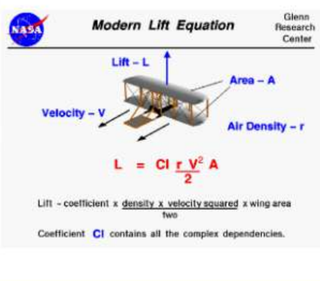


An Airplane's Angle of Attack

The angle of attack for an aircraft is the angle created between the inertial path and relative wind. An angle past the citical angle of attack (as show by the graph on the right) results in turbulant flow causing a dangerous loss of lift called stall.





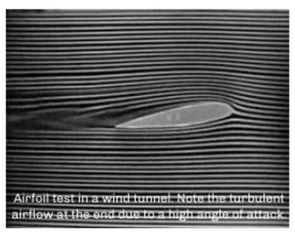


Lift Equation and Coefficient

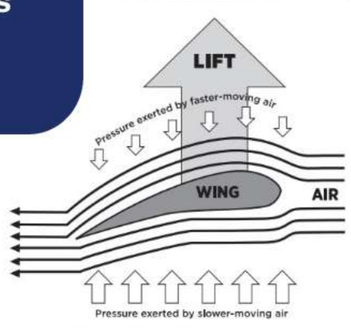
The lift equation is quite simple when you first think about it. Just find the values for your variables and plug it into the equation. But no... 🤔 The lift coefficient, Cl, takes into account several factors such as angle of attack and wing shape to help determine the magnitude of lift. Its most commonly determined by experimentation through wind tunnels, but can be determined through matematics as well.

Aerodynamic Lift

It is all about the airfoil... The camber, chord length, angle of attack, sharpness of the leading and trailing edge, stiffness, weight, and material.



Airfoil test in a wind tunnel. Note the turbulent airflow at the edge due to a high angle of attack.



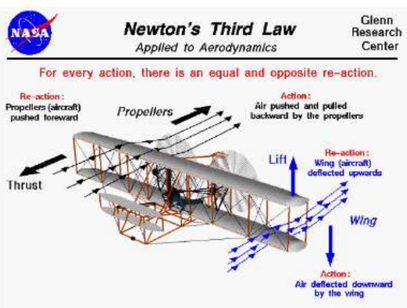
How is Lift Generated?

1 Bernoulli's Principle

"An increase in the speed of a fluid results in a corresponding decrease in its static pressure." The low velocity and high-pressure air on the bottom-side of an airfoil had the tendancy to move up towards the low pressure air creating lift.

2 Newton's 3rd Law

"For each force there is an equal and opposite reaction." The airfoil's shape deflects air downwards which lifts up the airplane.



Thrust & Drag

- Types of Engines
  - Most commonly found
- Forms of Drag
  - Parasite
  - Induced

DID YOU KNOW?

The Stratolaunch has 6 engines, 2 fuselages, and a 385-foot wingspan!

Types of Engines

1 Turbojet — ✗ Noisy ✗ Ineffective

One of the earliest gas turbine (AKA jet engine) is the turbojet, in which air passes through the compressor and combustion chamber. The hot air exits the engine and provides thrust for the plane.

2 Turbofan — ✓ Low Noise Level ✓ Effective @ Low Speeds

This is similar to the turbojet, however, there is a fan in front of the compressor which bypasses cool air around the engine for greater engine efficiency and significant noise reduction.

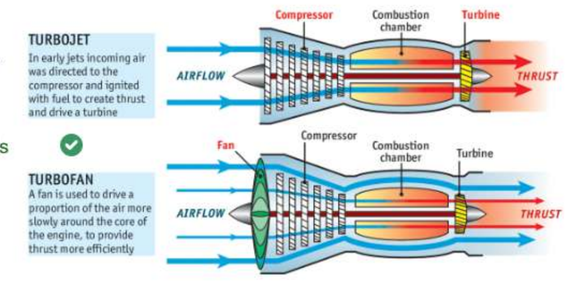
3 Turboprop — ✗ Not Durable ✓ Effective @ Mid Speeds and Altitudes

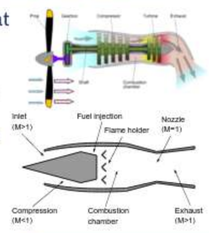
Essentially the turbojet with a geared propeller at the front for additional thrust.

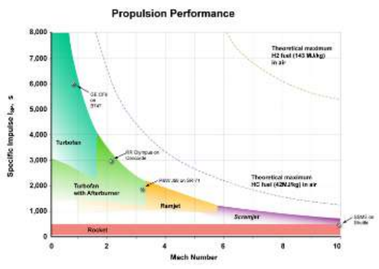
4 Ramjet — ✓ Effective at High Speeds

A supersonic engine which uses forward motion for air compression rather than a rotary compressor. Ineffective at subsonic speeds.

There are more engines, however these are the most common ones

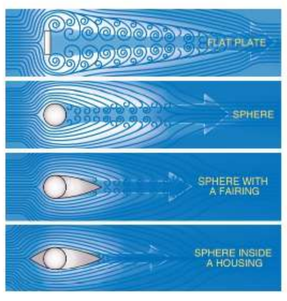






Piston Engines

Most commonly seen in general aviation aircraft, such as the Cessna 172. These are small and lightweight, allowing them to fit into small training and private aircraft. They are internal combustion piston engines just like those in cars. Such engines are easy to maintain and operate.



Types of Drag

Parasite Drag

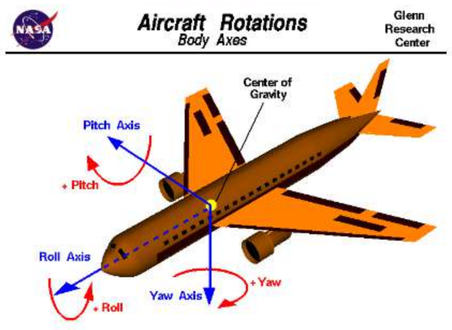
Forces that slow down an aircraft without any association with lift.

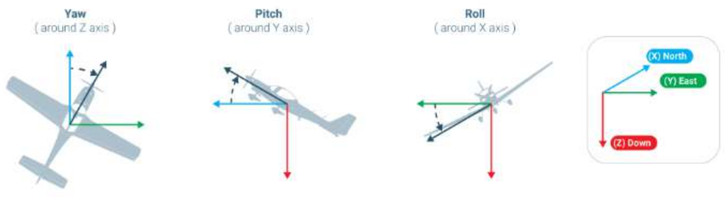
- Form Drag - generated by the aerodynamics of shapes on an aircraft. Extrusions such as engine cowlings and antennas can cause turbulent airflow, reducing thrust.
- Interference Drag - connections between different aircraft subsystems (i.e. the belly which connects the wing and fuselage) result in airflow collisions. This is worst at perpendicular angles, for which fairings should be used.
- Skin Friction Drag - air that contacts the roughness of a wing faces frication and turns into turbulent airflow. The extent to which this happens is determined by the skin material, roughness, compressibility, and stickiness.

Induced Drag

A required byproduct of lift - its needed in order for the airplane to have upward movement.

- Due to Bernoulli's Principle, the bottom-side high pressure wants to balance with the low pressure on the top of the airfoil, resulting in wing tip vortices. These inward spinning vortices can cause downwash which at times even help increase lift.





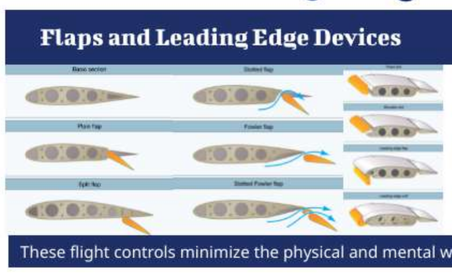
Control Surfaces

Primary and Secondary - they help control and navigate the aircraft during flight by rotating the aircraft on the x, y, and z axes.

General aviation planes usually use mechanical flight control systems. The control sticks and petals generally interact with cable pulleys and push rods to physically move control surfaces. Large commercial planes rely on complex hydraulics to move surfaces which have lots of extrenal pressure.

Secondary Flight Controls

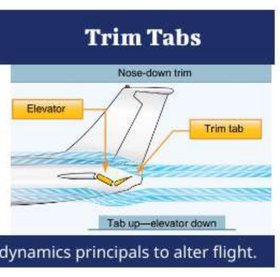
Flaps and Leading Edge Devices



Spoilers

Spoilers counteract the forces of lift and thrust. They're used for several purposes including decreasing ground roll while landing, counteracting adverse yaw during turns, and manipulating airflow around the aircraft for unpowered landings.

Trim Tabs



These flight controls minimize the physical and mental work pilots need to put in, by using aerodynamics principals to alter flight.

Pitch, up/down

Elevators
- Redirects air to pitch the aircraft up or down
- Can sometimes have a T-shape configuration to avoid propellor downwash
- Stabilator - one-piece horizontal stabilizer on a central hinge
- Canard - front horizontal stabilizer

Roll, right/left

Ailerons
- Present on each wing
- Deflects air up and down to roll right or left
- With ailerons deflected the side with more lift also has higher amounts of drag creating *adverse yaw* moving the plane opposite to the side of the turn - counteracted by use of the rudder

Yaw, right/left

Rudder
- Deflects airflow to turn right and left
- V-tail has such a shape to act as horizontal and vertical stabilizers
  - Use moveable called ruddervators which are connected via a linkage to allow simultaneous movement

Analog Instruments

Pitot-Static System

Pressure Data From: pitot tube (under the wing; pressure of air as airplane flies through it), static port (generally near fuselage; measures static pressure)

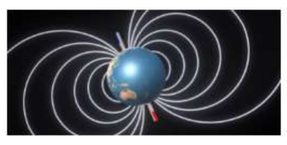
1. Altimeter (height) - static port; wafers which contract and expand based on pressure
2. Airspeed Indicator (speed) - pitot tube; expansion of the diaphragm connects with the dial through mechanical linkages
3. Vertical Speed Indicator (rate of climb/descent) - static port; comparing current pressure with previous pressure via the se of a diaphragm

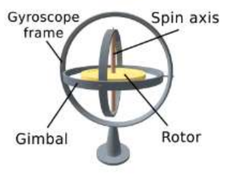
Magnetic Compass

2 magnets with a float in a liquid chamber

Variation (dependent on location in earth's surface) - difference between true and magnetic north

Deviation - compass error due to other magnetic objects within aircraft





Instruments & Navigation Systems

Gyroscopic Instruments

Data From: a gyroscope is a fast-spinning wheel which can freely rotate on several axes to provide reference direction

2 gyroscope principles: 1. rigidity in space (remain in a fixed position) and 2. precession (instead of rotation occurring at the point of a force, it occurs 90 degrees ahead of the direction of rotation)

1. Attitude Indicator (orientation relative to the horizon - pitch and roll) - air powered (by removing air)
2. Heading Indicator (yaw) - air powered (by removing air)
3. Turn Coordinator (roll backup) - electrically powered - comes with an inclinometer
   - Special conditions: slip (too little ruder) and skid (too much rudder)

Digital Displays

AKA Glass Cockpits

Garmin G1000

It integrates flight instruments, navigation menus, and communications into 2 displays: the primary flight display (basic flight instruments) and multi-function display (maps and system gauges).

How they work (2 components):
1. Air data computer (pitot-static system) - directly inputs data from the pitot static system and outputs the values using the computer
2. Attitude Heading Reference System (gyroscopic instruments) - use steady state sensors and a magnetometer

Using a Compass

Vocabulary:
1. Baseplate - holds the compass; has several lines and measurement
2. Rotating Bezel - a circle with 360-degree markings to easy identify the degree

Orient to North - turn the compass until the red line on the red marker (this may or may not adjust for magnetic north

Navigation – Basics

Orienting a Map

1. Place the compass on the map with the arrow up
2. Turn the bezel so that north is at the top
3. Rotate yourself until the north arrow is in the red box
4. Use the compass to identify landmarks near you - for reference, north is in front of you
5. Check the legend to understand what the map says

Sources | FAA Handbook | NASA | SKYbrary